(A) DATA CONSTRUCTION

1 Task & fields



- Maintain load
- Curtail loads
- Reduce output
- Suspend area

Plant energy manager

- Peak power
- Relief time
- Reduction
- Breach risk

2 Constraints

Cross-field constraints

Feasible ✓	Infeasible ×
304 ≤ 3,284	3,900 > 3,284

Dynamic constraints

Replacement	Peak → Risk		✓
Reversal	Relief time	+ → −	✓
Reversal	Reduction	− → +	×

3 Latent dynamics

Reduction	−	−	→ Old
Relief	+	−	
Peak	+	×	→ New
Risk	×	+	

Maintain
Curtail
Reduce
Suspend

4 Sampling & validation

Sample case

1,679 kW · 252 min
304 kW · 18%

Validation

Feasibility ✓ · Structural ✓
Numerical ✓ · Semantic ✓

✓ Qualified episodes

(B) RUNTIME PROTOCOL

1 Information

You are a plant energy manager.

Control electricity use.

Peak	-200 to 4,000 kW
Relief	0 to 480 min
Reduction	0 to 5,000 kW
Risk	0 to 100%

2 Proactive exploration

Probe Batch

1,679
252
304
18%

Calculator

Request Next Batch

3 Action selection

Evaluation batch

Case 1 … Case N

Submit Action Batch

4 Feedback & memory

Masked
Cases + actions

Kept
Utilities + probes

? Auxiliary assessment

Relevant fields?
Which directions?

- Peak power
- More intervention

(C) EXAMPLE TRAJECTORY

1 Probe case

Probe Batch

Peak	1,679 kW
Relief	252 min
Reduction	304 kW
Breach risk	18%

2 Before change

Rule inferred

Maintain
Curtail
Reduce
Suspend

3 Dynamic change

Performance drop

4 Re-probe

New evidence

Maintain
Curtail
Reduce
Suspend

5 Adaptation outcome

✓ Adapted

Uncertain